

Circles & Coordinate Geometry

ANSWER KEY



The equation of a circle

- 1 Write the equation of a circle with center $(-4, 7)$ and radius 9.

$$(x + 4)^2 + (y - 7)^2 = 81.$$

- 2 The circle below has the radius shown. Find its area and circumference, in terms of π .

$$\text{Area} = \pi(6)^2 = 36\pi. \text{ Circumference} = 2\pi(6) = 12\pi.$$

- 3 A circle has diameter 26 and is centered at the origin. Write its equation, and state its radius.

$$\text{Radius} = \frac{26}{2} = 13. \text{ Equation: } x^2 + y^2 = 169.$$

Completing the square

- 4 Find the center and radius of $x^2 + y^2 - 6x + 8y - 11 = 0$.

$$(x^2 - 6x + 9) + (y^2 + 8y + 16) = 11 + 9 + 16 = 36 \Rightarrow (x - 3)^2 + (y + 4)^2 = 36. \text{ Center } (3, -4), \text{ radius } 6.$$

- 5 Find the center and radius of $x^2 + y^2 + 10x - 4y + 13 = 0$.

$$(x^2 + 10x + 25) + (y^2 - 4y + 4) = -13 + 25 + 4 = 16 \Rightarrow (x + 5)^2 + (y - 2)^2 = 16. \text{ Center } (-5, 2), \text{ radius } 4.$$

Distance, midpoint, and slope

- 6 Find the distance between $(-3, 5)$ and $(4, -7)$, to the nearest hundredth.

$$d = \sqrt{(4 - (-3))^2 + (-7 - 5)^2} = \sqrt{49 + 144} = \sqrt{193} \approx 13.89.$$

- 7 Find the midpoint and slope of the segment joining $(2, -6)$ and $(-8, 10)$.

$$\text{Midpoint} = \left(\frac{2 - 8}{2}, \frac{-6 + 10}{2} \right) = (-3, 2). \text{ Slope} = \frac{10 - (-6)}{-8 - 2} = \frac{16}{-10} = -\frac{8}{5}.$$

- 8 Find the slope of the line through $(-2, 7)$ and $(6, -5)$, and state whether the line rises or falls from left to right.

$$\text{Slope} = \frac{-5 - 7}{6 - (-2)} = \frac{-12}{8} = -\frac{3}{2}. \text{ Since the slope is negative, the line falls from left to right.}$$

Coordinate proofs

- 9 Show that the triangle with vertices $A(0, 0)$, $B(4, 0)$, $C(4, 3)$ is a right triangle, using slopes.

Slope of $AB = \frac{0-0}{4-0} = 0$ (horizontal). Slope of BC is undefined (vertical, since $x = 4$ for both points). A horizontal segment and a vertical segment are perpendicular, so $\angle B = 90^\circ$, confirming a right triangle.

- 10 Show that $A(1, 2)$, $B(4, 3)$, $C(3, 6)$, $D(0, 5)$ form a parallelogram, by showing both pairs of opposite sides are parallel.

Slope $AB = \frac{3-2}{4-1} = \frac{1}{3}$. Slope $DC = \frac{6-5}{3-0} = \frac{1}{3}$, so $AB \parallel DC$. Slope $AD = \frac{5-2}{0-1} = -3$. Slope $BC = \frac{6-3}{3-4} = -3$, so $AD \parallel BC$. Both pairs of opposite sides are parallel, so $ABCD$ is a parallelogram.

Tangent lines to circles

- 11 A circle has center $O(2, 3)$ and radius 5. Point $P(2, 12)$ lies outside the circle. Find the length of the tangent from P to the circle, to the nearest hundredth.

$$OP = \sqrt{(2-2)^2 + (12-3)^2} = \sqrt{81} = 9. \text{ Tangent length} \\ = \sqrt{OP^2 - r^2} = \sqrt{81 - 25} = \sqrt{56} = 2\sqrt{14} \approx 7.48.$$

- 12 Find the equation of the tangent line to the circle $x^2 + y^2 = 25$ at the point $(3, 4)$ on the circle.

The radius to $(3, 4)$ has slope $\frac{4}{3}$, so the tangent (perpendicular to the radius) has slope $-\frac{3}{4}$. Line: $y - 4 = -\frac{3}{4}(x - 3)$, which simplifies to $3x + 4y = 25$.

- 13 A circle has center $(0, 0)$ and radius 10. Determine whether the point $(6, 9)$ lies inside, on, or outside the circle.

Distance from center $= \sqrt{6^2 + 9^2} = \sqrt{117} \approx 10.82$, which is greater than the radius 10. The point lies outside the circle.

Circle theorems

- 14 A central angle of 84° subtends an arc. Find the measure of the inscribed angle subtending the same arc.

$$\text{Inscribed angle} = \frac{1}{2}(84^\circ) = 42^\circ.$$

- 15 A chord subtends the central angle shown below. Find the measure of the inscribed angle subtending the same (minor) arc.

$$\text{Inscribed angle} = \frac{1}{2}(100^\circ) = 50^\circ.$$

Arc length and sector area

- 16 A circle has radius 12 cm. Find the arc length subtended by a central angle of $\frac{5\pi}{6}$ radians.

$$s = r\theta = 12\left(\frac{5\pi}{6}\right) = 10\pi \text{ cm.}$$

- 17 The circle below has the radius and shaded sector shown. Find the area of the sector, in terms of π .

$$72^\circ = \frac{2\pi}{5} \text{ radians. Sector area} = \frac{1}{2}r^2\theta = \frac{1}{2}(100)\left(\frac{2\pi}{5}\right) = 20\pi.$$

- 18 A sector of a circle with radius 9 has area 27π . Find the central angle, in radians.

$$27\pi = \frac{1}{2}(81)\theta = 40.5\theta \Rightarrow \theta = \frac{27\pi}{40.5} = \frac{2\pi}{3}.$$

Conics: ellipses and hyperbolas

- 19 Identify the type of conic and state its center for $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$.

An ellipse, centered at $(2, -1)$, with $a^2 = 16$ ($a = 4$, horizontal major axis) and $b^2 = 9$ ($b = 3$).

- 20 Identify the type of conic, its center, and its asymptote slopes for $\frac{(x+3)^2}{4} - \frac{y^2}{9} = 1$.

A hyperbola, centered at $(-3, 0)$, with $a = 2$, $b = 3$. Asymptote slopes: $\pm \frac{b}{a} = \pm \frac{3}{2}$.

A well-designed word problem: a Ferris wheel

- 21 This question has three parts. A circular Ferris wheel has a diameter of 40 m, with its center 22 m above the ground. Set up a coordinate system with the origin at ground level, directly below the center. (a) Write the equation of the circle traced by a passenger cabin. (b) A support cable runs from a ground anchor at $(26, 0)$ to a point of tangency on the wheel. Find the length of this cable, to the nearest hundredth of a meter. (c) Find the height above the ground of a cabin when it is 12 m horizontally from the vertical line through the center.

(a) Center $(0, 22)$, radius 20: $x^2 + (y - 22)^2 = 400$. (b) $OP = \sqrt{26^2 + 22^2} = \sqrt{1160} \approx 34.06$ m.

Tangent length $= \sqrt{OP^2 - r^2} = \sqrt{1160 - 400} = \sqrt{760} \approx 27.57$ m. (c) With $x = 12$:

$$12^2 + (y - 22)^2 = 400 \Rightarrow (y - 22)^2 = 256 \Rightarrow y = 22 \pm 16, \text{ so } y = 38 \text{ m or } y = 6 \text{ m.}$$

Multiple choice — easy

22 Write the equation of a circle with center (9, 12) and radius 12.

a. $(x - 9)^2 + (y - 12)^2 = 12$

b. $(x + 9)^2 + (y + 12)^2 = 144$

c. $(x - 9)^2 + (y + 12)^2 = 144$

d. $(x - 9)^2 + (y - 12)^2 = 144$ ✓

$(x - 9)^2 + (y - 12)^2 = 144.$

23 Write the equation of a circle with center (4, 2) and radius 2.

a. $(x + 4)^2 + (y + 2)^2 = 4$

b. $(x - 4)^2 + (y + 2)^2 = 4$

c. $(x - 4)^2 + (y - 2)^2 = 2$

d. $(x - 4)^2 + (y - 2)^2 = 4$ ✓

$(x - 4)^2 + (y - 2)^2 = 4.$

24 Write the equation of a circle with center (8, -7) and radius 7.

a. $(x - 8)^2 + (y - 7)^2 = 49$

b. $(x + 8)^2 + (y - 7)^2 = 49$

c. $(x - 8)^2 + (y + 7)^2 = 7$

d. $(x - 8)^2 + (y + 7)^2 = 49$ ✓

$(x - 8)^2 + (y + 7)^2 = 49.$

25 Write the equation of a circle with center (-6, 2) and radius 6.

a. $(x - 6)^2 + (y + 2)^2 = 36$

b. $(x + 6)^2 + (y - 2)^2 = 36$ ✓

c. $(x + 6)^2 + (y + 2)^2 = 36$

d. $(x + 6)^2 + (y - 2)^2 = 6$

$(x + 6)^2 + (y - 2)^2 = 36.$

26 Write the equation of a circle with center $(11, -9)$ and radius 5.

a. $(x + 11)^2 + (y - 9)^2 = 25$

b. $(x - 11)^2 + (y + 9)^2 = 25$ ✓

c. $(x - 11)^2 + (y + 9)^2 = 5$

d. $(x - 11)^2 + (y - 9)^2 = 25$

$(x - 11)^2 + (y + 9)^2 = 25.$

27 Write the equation of a circle with center $(10, -2)$ and radius 10.

a. $(x + 10)^2 + (y - 2)^2 = 100$

b. $(x - 10)^2 + (y - 2)^2 = 100$

c. $(x - 10)^2 + (y + 2)^2 = 10$

d. $(x - 10)^2 + (y + 2)^2 = 100$ ✓

$(x - 10)^2 + (y + 2)^2 = 100.$

28 Write the equation of a circle with center $(0, 10)$ and radius 10.

a. $(x)^2 + (y - 10)^2 = 102$

b. $(x)^2 + (y - 10)^2 = 100$ ✓

c. $(x)^2 + (y - 10)^2 = 10$

d. $(x)^2 + (y + 10)^2 = 100$

$(x)^2 + (y - 10)^2 = 100.$

29 Write the equation of a circle with center $(-11, -3)$ and radius 13.

a. $(x + 11)^2 + (y - 3)^2 = 169$

b. $(x + 11)^2 + (y + 3)^2 = 13$

c. $(x + 11)^2 + (y + 3)^2 = 169$ ✓

d. $(x - 11)^2 + (y - 3)^2 = 169$

$(x + 11)^2 + (y + 3)^2 = 169.$

30 Write the equation of a circle with center $(-5, 10)$ and radius 4.

a. $(x - 5)^2 + (y + 10)^2 = 16$

b. $(x + 5)^2 + (y - 10)^2 = 16$ ✓

c. $(x + 5)^2 + (y - 10)^2 = 4$

d. $(x + 5)^2 + (y + 10)^2 = 16$

$(x + 5)^2 + (y - 10)^2 = 16.$

31 Write the equation of a circle with center $(2, 5)$ and radius 12.

a. $(x - 2)^2 + (y - 5)^2 = 12$

b. $(x - 2)^2 + (y + 5)^2 = 144$

c. $(x + 2)^2 + (y + 5)^2 = 144$

d. $(x - 2)^2 + (y - 5)^2 = 144$ ✓

$(x - 2)^2 + (y - 5)^2 = 144.$

32 Write the equation of a circle with center $(-2, 1)$ and radius 3.

a. $(x + 2)^2 + (y + 1)^2 = 9$

b. $(x + 2)^2 + (y - 1)^2 = 9$ ✓

c. $(x - 2)^2 + (y + 1)^2 = 9$

d. $(x + 2)^2 + (y - 1)^2 = 3$

$(x + 2)^2 + (y - 1)^2 = 9.$

33 Write the equation of a circle with center $(-1, 9)$ and radius 11.

a. $(x + 1)^2 + (y - 9)^2 = 121$ ✓

b. $(x - 1)^2 + (y + 9)^2 = 121$

c. $(x + 1)^2 + (y + 9)^2 = 121$

d. $(x + 1)^2 + (y - 9)^2 = 11$

$(x + 1)^2 + (y - 9)^2 = 121.$

34 Write the equation of a circle with center $(-6, 6)$ and radius 13.

a. $(x + 6)^2 + (y + 6)^2 = 169$

b. $(x + 6)^2 + (y - 6)^2 = 169$ ✓

c. $(x - 6)^2 + (y + 6)^2 = 169$

d. $(x + 6)^2 + (y - 6)^2 = 13$

$(x + 6)^2 + (y - 6)^2 = 169.$

35 Write the equation of a circle with center $(11, -5)$ and radius 13.

a. $(x + 11)^2 + (y - 5)^2 = 169$

b. $(x - 11)^2 + (y + 5)^2 = 13$

c. $(x - 11)^2 + (y - 5)^2 = 169$

d. $(x - 11)^2 + (y + 5)^2 = 169$ ✓

$(x - 11)^2 + (y + 5)^2 = 169.$

36 Write the equation of a circle with center $(-7, 2)$ and radius 7.

a. $(x + 7)^2 + (y - 2)^2 = 49$ ✓

b. $(x - 7)^2 + (y + 2)^2 = 49$

c. $(x + 7)^2 + (y + 2)^2 = 49$

d. $(x + 7)^2 + (y - 2)^2 = 7$

$(x + 7)^2 + (y - 2)^2 = 49.$

Multiple choice — medium

37 Find the center and radius of $x^2 + y^2 + 2x - 2y - 62 = 0$.

a. $(-1, -1), 8$

b. $(-1, 1), 8$ ✓

c. $(1, 1), 8$

d. $(-1, 1), 10$

$(x^2 + 2x + 1) + (y^2 - 2y + 1) = 62 + 1 + 1 \Rightarrow (x + 1)^2 + (y - 1)^2 = 64$. Center $(-1, 1)$, radius 8.

38 Find the center and radius of $x^2 + y^2 - 16x + 2y - 79 = 0$.

a. $(8, 1), 12$

b. $(-8, -1), 12$

c. $(8, -1), 14$

d. $(8, -1), 12$ ✓

$(x^2 - 16x + 64) + (y^2 + 2y + 1) = 79 + 64 + 1 \Rightarrow (x - 8)^2 + (y + 1)^2 = 144$. Center $(8, -1)$, radius 12.

39 Find the center and radius of $x^2 + y^2 + 12x - 6y - 99 = 0$.

a. $(-6, 3), 14$

b. $(-6, 3), 12$ ✓

c. $(-6, -3), 12$

d. $(6, 3), 12$

$(x^2 + 12x + 36) + (y^2 - 6y + 9) = 99 + 36 + 9 \Rightarrow (x + 6)^2 + (y - 3)^2 = 144$. Center $(-6, 3)$, radius 12.

40 Find the center and radius of $x^2 + y^2 - 6x - 2y - 90 = 0$.

a. $(3, 1), 12$

b. $(-3, 1), 10$

c. $(3, 1), 10$ ✓

d. $(3, -1), 10$

$(x^2 - 6x + 9) + (y^2 - 2y + 1) = 90 + 9 + 1 \Rightarrow (x - 3)^2 + (y - 1)^2 = 100$. Center $(3, 1)$, radius 10.

41 Find the center and radius of $x^2 + y^2 - 16x + 12y - 21 = 0$.

a. $(8, 6), 11$

b. $(-8, -6), 11$

c. $(8, -6), 11$ ✓

d. $(8, -6), 13$

$(x^2 - 16x + 64) + (y^2 + 12y + 36) = 21 + 64 + 36 \Rightarrow (x - 8)^2 + (y + 6)^2 = 121$. Center $(8, -6)$, radius 11.

42 Find the center and radius of $x^2 + y^2 - 8x - 18y + 81 = 0$.

- a. $(-4, 9), 4$
- b. $(4, -9), 4$
- c. $(4, 9), 6$
- d. $(4, 9), 4$ ✓

$(x^2 - 8x + 16) + (y^2 - 18y + 81) = -81 + 16 + 81 \Rightarrow (x - 4)^2 + (y - 9)^2 = 16$. Center $(4, 9)$, radius 4.

43 Find the center and radius of $x^2 + y^2 - 14x - 8y + 1 = 0$.

- a. $(7, 4), 10$
- b. $(7, 4), 8$ ✓
- c. $(7, -4), 8$
- d. $(-7, 4), 8$

$(x^2 - 14x + 49) + (y^2 - 8y + 16) = -1 + 49 + 16 \Rightarrow (x - 7)^2 + (y - 4)^2 = 64$. Center $(7, 4)$, radius 8.

44 Find the center and radius of $x^2 + y^2 - 10x + 8y + 37 = 0$.

- a. $(5, 4), 2$
- b. $(-5, -4), 2$
- c. $(5, -4), 2$ ✓
- d. $(5, -4), 4$

$(x^2 - 10x + 25) + (y^2 + 8y + 16) = -37 + 25 + 16 \Rightarrow (x - 5)^2 + (y + 4)^2 = 4$. Center $(5, -4)$, radius 2.

45 Find the center and radius of $x^2 + y^2 - 14x + 2y - 50 = 0$.

a. $(7, -1), 10$ ✓

b. $(7, -1), 12$

c. $(7, 1), 10$

d. $(-7, -1), 10$

$(x^2 - 14x + 49) + (y^2 + 2y + 1) = 50 + 49 + 1 \Rightarrow (x - 7)^2 + (y + 1)^2 = 100$. Center $(7, -1)$, radius 10.

46 Find the center and radius of $x^2 + y^2 - 12x - 2y - 84 = 0$.

a. $(6, 1), 13$

b. $(-6, 1), 11$

c. $(6, 1), 11$ ✓

d. $(6, -1), 11$

$(x^2 - 12x + 36) + (y^2 - 2y + 1) = 84 + 36 + 1 \Rightarrow (x - 6)^2 + (y - 1)^2 = 121$. Center $(6, 1)$, radius 11.

47 Find the center and radius of $x^2 + y^2 + 4x + 14y - 91 = 0$.

a. $(-2, -7), 12$ ✓

b. $(2, -7), 12$

c. $(-2, -7), 14$

d. $(-2, 7), 12$

$(x^2 + 4x + 4) + (y^2 + 14y + 49) = 91 + 4 + 49 \Rightarrow (x + 2)^2 + (y + 7)^2 = 144$. Center $(-2, -7)$, radius 12.

48 Find the center and radius of $x^2 + y^2 - 16x - 2y + 56 = 0$.

a. $(8, -1), 3$

b. $(8, 1), 3$ ✓

c. $(-8, 1), 3$

d. $(8, 1), 5$

$(x^2 - 16x + 64) + (y^2 - 2y + 1) = -56 + 64 + 1 \Rightarrow (x - 8)^2 + (y - 1)^2 = 9$. Center $(8, 1)$, radius 3.

49 Find the center and radius of $x^2 + y^2 + 12x - 18y + 17 = 0$.

a. $(-6, 9), 12$

b. $(6, 9), 10$

c. $(-6, 9), 10$ ✓

d. $(-6, -9), 10$

$(x^2 + 12x + 36) + (y^2 - 18y + 81) = -17 + 36 + 81 \Rightarrow (x + 6)^2 + (y - 9)^2 = 100$. Center $(-6, 9)$, radius 10.

50 Find the center and radius of $x^2 + y^2 - 8y - 33 = 0$.

a. $(0, -4), 7$

b. $(2, 4), 7$

c. $(0, 4), 9$

d. $(0, 4), 7$ ✓

$(x^2) + (y^2 - 8y + 16) = 33 + 16 \Rightarrow (x)^2 + (y - 4)^2 = 49$. Center $(0, 4)$, radius 7.

51 Find the center and radius of $x^2 + y^2 - 2x + 4y - 116 = 0$.

- a. $(1, -2), 13$
- b. $(-1, -2), 11$
- c. $(1, 2), 11$
- d. $(1, -2), 11$ ✓

$$(x^2 - 2x + 1) + (y^2 + 4y + 4) = 116 + 1 + 4 \Rightarrow (x - 1)^2 + (y + 2)^2 = 121. \text{ Center } (1, -2), \text{ radius } 11.$$

Multiple choice — hard

52 A circle has center $O(8, 4)$ and radius 20. Point $P(8, 33)$ lies outside the circle. Find the length of the tangent from P to the circle.

- a. 29
- b. 23
- c. 21 ✓
- d. 19

$$OP = \sqrt{(8 - 8)^2 + (33 - 4)^2} = \sqrt{841} = 29. \text{ Tangent length} \\ = \sqrt{OP^2 - r^2} = \sqrt{841 - 400} = \sqrt{441} = 21.$$

53 A circle has center $O(-6, 7)$ and radius 5. Point $P(-6, 20)$ lies outside the circle. Find the length of the tangent from P to the circle.

- a. 13
- b. 14
- c. 12 ✓
- d. 10

$$OP = \sqrt{((-6) - (-6))^2 + (20 - 7)^2} = \sqrt{169} = 13. \text{ Tangent length} \\ = \sqrt{OP^2 - r^2} = \sqrt{169 - 25} = \sqrt{144} = 12.$$

- 54 A circle has center $O(1, 1)$ and radius 20. Point $P(1, 30)$ lies outside the circle. Find the length of the tangent from P to the circle.

a. 21 ✓

b. 23

c. 29

d. 19

$$OP = \sqrt{(1-1)^2 + (30-1)^2} = \sqrt{841} = 29. \text{ Tangent length} \\ = \sqrt{OP^2 - r^2} = \sqrt{841 - 400} = \sqrt{441} = 21.$$

- 55 A circle has center $O(-4, -3)$ and radius 7. Point $P(-4, 22)$ lies outside the circle. Find the length of the tangent from P to the circle.

a. 22

b. 24 ✓

c. 25

d. 26

$$OP = \sqrt{((-4)-(-4))^2 + (22-(-3))^2} = \sqrt{625} = 25. \text{ Tangent length} \\ = \sqrt{OP^2 - r^2} = \sqrt{625 - 49} = \sqrt{576} = 24.$$

- 56 A circle has center $O(7, 5)$ and radius 5. Point $P(7, 18)$ lies outside the circle. Find the length of the tangent from P to the circle.

a. 13

b. 12 ✓

c. 10

d. 14

$$OP = \sqrt{(7-7)^2 + (18-5)^2} = \sqrt{169} = 13. \text{ Tangent length} = \sqrt{OP^2 - r^2} = \sqrt{169 - 25} = \sqrt{144} = 12.$$

- 57 A circle has radius 8. Find the area of the sector with central angle 60° , in terms of π .

- a. $\frac{29}{3}\pi$
- b. $\frac{32}{3}\pi$ ✓
- c. $\frac{64}{3}\pi$
- d. $\frac{35}{3}\pi$

$$\text{Sector area} = \frac{60}{360} \times \pi(8)^2 = \frac{60}{360} \times 64\pi = \frac{32}{3}\pi.$$

- 58 A circle has radius 13. Find the area of the sector with central angle 40° , in terms of π .

- a. $\frac{338}{9}\pi$
- b. $\frac{178}{9}\pi$
- c. $\frac{160}{9}\pi$
- d. $\frac{169}{9}\pi$ ✓

$$\text{Sector area} = \frac{40}{360} \times \pi(13)^2 = \frac{40}{360} \times 169\pi = \frac{169}{9}\pi.$$

- 59 A circle has radius 4. Find the area of the sector with central angle 36° , in terms of π .

- a. $\frac{8}{5}\pi$ ✓
- b. $\frac{3}{5}\pi$
- c. $\frac{13}{5}\pi$
- d. $\frac{16}{5}\pi$

$$\text{Sector area} = \frac{36}{360} \times \pi(4)^2 = \frac{36}{360} \times 16\pi = \frac{8}{5}\pi.$$

- 60 A circle has radius 12. Find the area of the sector with central angle 150° , in terms of π .

- a. 60π ✓
- b. 61π
- c. 120π
- d. 59π

$$\text{Sector area} = \frac{150}{360} \times \pi(12)^2 = \frac{150}{360} \times 144\pi = 60\pi.$$

61 A circle has radius 14. Find the area of the sector with central angle 240° , in terms of π .

a. $\frac{392}{3}\pi$ ✓

b. $\frac{784}{3}\pi$

c. $\frac{389}{3}\pi$

d. $\frac{395}{3}\pi$

$$\text{Sector area} = \frac{240}{360} \times \pi(14)^2 = \frac{240}{360} \times 196\pi = \frac{392}{3}\pi.$$

62 Find the distance between $(1, 1)$ and $(-1, 9)$, to the nearest hundredth.

a. 7.25

b. 8.25 ✓

c. 10.25

d. 9.25

$$d = \sqrt{((-1) - 1)^2 + (9 - 1)^2} = \sqrt{4 + 64} = \sqrt{68} \approx 8.25.$$

63 Find the distance between $(0, -7)$ and $(-1, 1)$, to the nearest hundredth.

a. 9.06

b. 7.06

c. 8.06 ✓

d. 10.06

$$d = \sqrt{((-1) - 0)^2 + (1 - (-7))^2} = \sqrt{1 + 64} = \sqrt{65} \approx 8.06.$$

64 Find the distance between $(9, -6)$ and $(6, 9)$, to the nearest hundredth.

- a. 14.30
- b. 17.30
- c. 15.30 ✓
- d. 16.30

$$d = \sqrt{(6 - 9)^2 + (9 - (-6))^2} = \sqrt{9 + 225} = \sqrt{234} \approx 15.30.$$

65 Find the distance between $(5, 9)$ and $(-2, 1)$, to the nearest hundredth.

- a. 9.63
- b. 12.63
- c. 10.63 ✓
- d. 11.63

$$d = \sqrt{((-2) - 5)^2 + (1 - 9)^2} = \sqrt{49 + 64} = \sqrt{113} \approx 10.63.$$

66 Find the distance between $(-3, 10)$ and $(2, -3)$, to the nearest hundredth.

- a. 12.93
- b. 14.93
- c. 15.93
- d. 13.93 ✓

$$d = \sqrt{(2 - (-3))^2 + ((-3) - 10)^2} = \sqrt{25 + 169} = \sqrt{194} \approx 13.93.$$